DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE COMPUTADORES E INFORMÁTICA

PRACTICAL GUIDE NO. 7

Task 1 - Simulation of video-streaming server

Consider the provided Matlab function `VideoStreamingSimulator()` that implements a discrete event simulator of a video-streaming server. Save the file to your Matlab current folder. A description of the input parameters is obtained running in the Matlab Command Window:

```
>> help VideoStreamingSimulator

  VideoStreamingSimulator(lambda, invmiu, B, M, N)
  lambda: arrival rate of movie requests (requests/minute)
  invmiu: average duration of movies (minutes)
  B:      throughput of each movie (Mbps)
  M:      capacity of the server (no. of simultaneous transmitted movies)
  N:      stopping criterion (the simulation finishes at the end of the
          transmition of the Nth movie)
```

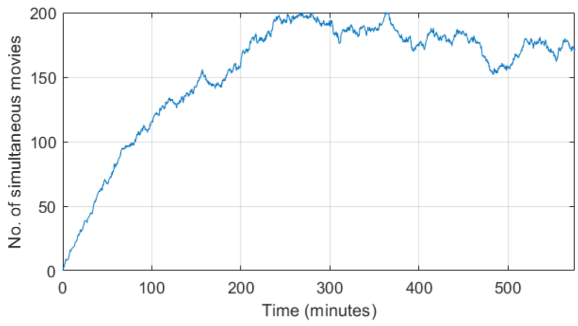
The simulator assumes that the arrival of movie requests is a Poisson process and the movie duration is exponentially distributed.

1. Based on the simulator provided, develop a new function to visualize the evolution of the number of simultaneous movies transmitted by the server during the simulation run. Then, consider an average movie duration of 90 minutes, B = 2 Mbps and M = 200, and run the simulator for the cases:

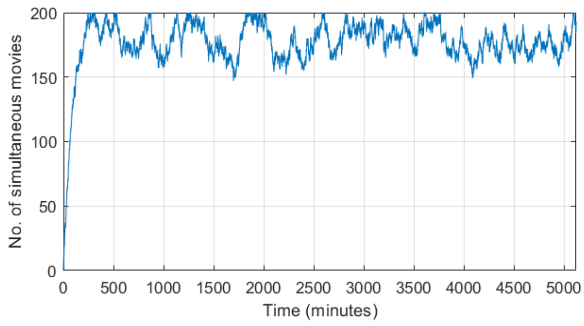
(a) arrival rate of 2 requests/minute and $N = 10^3$,
(b) arrival rate of 2 requests/minute and $N = 10^4$,
(c) arrival rate of 3 requests/minute and $N = 10^3$,
(d) arrival rate of 3 requests/minute and $N = 10^4$.

Justify the visualization differences in the observed initial transient state of the simulation runs before reaching the stationary state. The visualizations should look like:

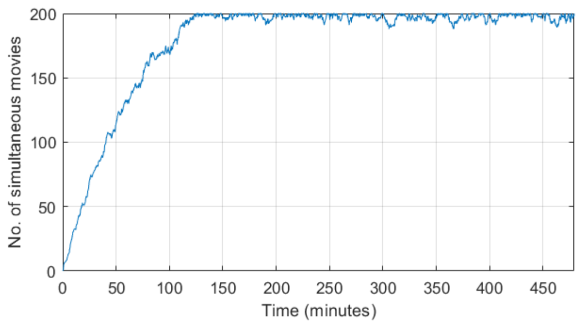
(a)



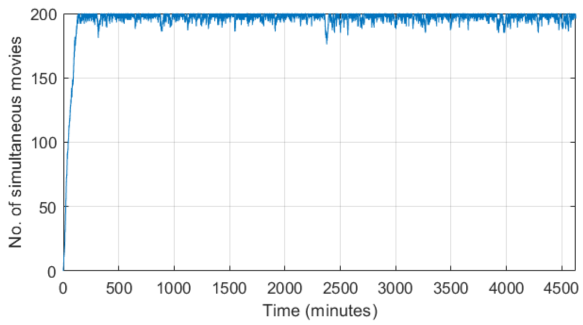
(b)



(c)



(d)



2. Based on the simulator provided, develop a new simulator to add the estimation of the following two performance parameters:

PB – probability of a movie request being blocked (i.e., being refused) because the server is transmitting at its maximum capacity (in number of simultaneous movies)
DB – average throughput (in Mbps) of the movies being transmitted by the server

Then, consider the case of an arrival rate of 2 requests/minute, an average movie duration of 90 minutes, B = 2 Mbps, M = 200 and $N = 10^4$. Compute the estimated values and the associated 90% confidence intervals of the performance parameters PB and DB:

(a) based on 10 runs of this simulator,

(b) based on 100 runs of this simulator.

Compare the width of the obtained confidence intervals of the two cases and draw conclusions. The results should look like:

```
Exercise 2(a):
   PB = 0.0097 [0.0076 - 0.0117]
   DB = 350.21 Mbps [348.56 - 351.87]
Exercise 2(b):
   PB = 0.0103 [0.0096 - 0.0110]
   DB = 350.76 Mbps [350.06 - 351.45]
```

3. So far, the simulations have considered an exponentially distributed movie duration. Now, the aim is to consider the movie duration statistics based on some input data.

First, compute the parameters of a discrete variable representing the movie duration statistics based on the provided CSV file `moviesMPECI2025.csv` and using the method proposed in TP classes. Register the resulting average movie duration D and plot the probability mass function of the discrete variable.

Then, based on the simulator developed in **2.**, develop a new simulator changing the statistics of the movie duration to the ones of the previous discrete variable.

Finally, consider the case of an arrival rate of 2 requests/minute, B = 2 Mbps, M = 200 and $N = 10^5$. Compute the estimated values and the associated 90% confidence intervals of the performance parameters PB and DB based on 20 runs using:

(a) this simulator,

(b) the simulator developed in **2.** with an average movie duration of D minutes.

Compare the results of the two cases and draw conclusions. The results should look like:

```
Exercise 3(a):
   PB = 0.0567 [0.0561 - 0.0574]
   DB = 378.60 Mbps [378.46 - 378.75]
Exercise 3(b):
   PB = 0.0561 [0.0552 - 0.0570]
   DB = 378.00 Mbps [377.67 - 378.32]
```

4. Consider an arrival rate of 2 requests/minute, an average movie duration of D (computed in **3.**), B = 2 Mbps and M = 200. Determine the theoretical values of the performance parameters PB and DB by modelling the video-streaming server as a $M/M/m/m$ queuing system. Compare these results with the previous simulation results (obtained in **3.)** and draw conclusions.

Task 2 - Simulation of a data link connection

Consider the provided Matlab function `LinkSimulator()` that implements a discrete event simulator of a data link connection. Save the file to your Matlab current folder. A description of the input and output parameters is obtained running in the Matlab Command Window:

```
>> help LinkSimulator

  AM = LinkSimulator(lambda,B,C,F,N)

  Input:
      lambda: packet arrival rate (in packets/second)
      B:      av.g packet size (in Bytes)
      C:      link capacity (in Mbps)
      F:      queue capacity (in packets)
      N:      stopping criterion (the simulation finishes at the end of the
              transmition of the Nth packet)
  Output:
      AM:     average packet delay (in seconds)
```

The simulator assumes that the arrival of packets is a Poisson process and the packet size is exponentially distributed. The simulator includes the appropriate statistical counters to estimate at the end the AM (average packet delay) performance parameter.

5. Consider the case of a packet arrival rate of 1000 pps (packets/second), an average packet size of 600 Bytes, a link capacity of 10 Mbps (Megabits/second) and a queue capacity of 1000 packets. Compute the estimated value and the associated 90% confidence interval of AM (in milliseconds), based on 100 runs of the simulator with $N = 10^4$. The results should look like:

```
Exercise 5:
   AM = 0.9231 [0.9191 - 0.9271] msec
```

6. The provided simulator assumes that the packet size is exponentially distributed. Based on the simulator provided, develop a new simulator changing the statistics of the packet size to consider that the size of each packet is between 64 and 1500 Bytes with the probabilities:

– 64 Bytes with probability 29.606%,

– 110 Bytes with probability 16.417%,

– 1500 Bytes with probability 19.601%,

– all other values (i.e., from 65 to 109 and from 111 to 1500) with the same probability.

Check that the resulting average packet size is 600 Bytes (as considered in **5.**).

Then, consider again the case of a packet arrival rate of 1000 pps, a link capacity of 10 Mbps and a queue capacity of 1000 packets. Compute the estimated value and the associated 90% confidence interval of AM (in milliseconds), based on 100 runs of the simulator with $N = 10^4$. Compare these results with the previous ones (obtained in **5.**) and draw conclusions. The results should look like:

```
Exercise 6:
   Avg. Packet Size = 600.0030
   AM = 0.9190 [0.9152 - 0.9228] msec
```

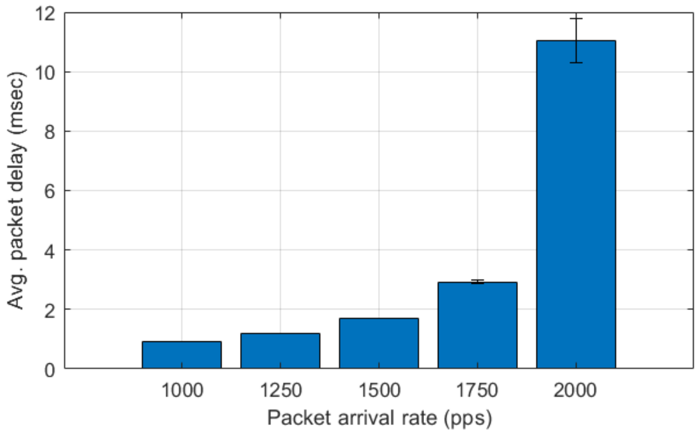
7. Consider the case of a link capacity of 10 Mbps and a queue capacity of 1000 packets. With the simulator developed in **6.**, compute the estimated value and the associated 90% confidence interval of AM based on 100 runs of the simulator with $N = 10^4$ considering:

(a) a packet arrival rate of 1250 pps,
(b) a packet arrival rate of 1500 pps,
(c) a packet arrival rate of 1750 pps,
(d) a packet arrival rate of 2000 pps.

Compare these results and the previous result (obtained in **6.**) and draw conclusions. The results should look like:

```
Exercise 7:
   lambda= 1250: AM = 1.1950 [1.1878 - 1.2021] msec
   lambda= 1500: AM = 1.6996 [1.6839 - 1.7153] msec
   lambda= 1750: AM = 2.9288 [2.8791 - 2.9784] msec
   lambda= 2000: AM = 11.0441 [10.3110 - 11.7772] msec
```

8. Present the simulation results of **6.** and **7.** in a bar chart with the confidence intervals in error bars on the same plot [Footnote 1]. Analyse the width of the obtained confidence intervals and take conclusions on the confidence of the results. The visualization should look like:



9. Based on the simulator developed in **6.**, develop a new simulator to add the estimation of the following performance parameter:

PD – packet drop (i.e., the probability of a packet being discarded because the queue is full when it arrives)

Then, consider the case of a packet arrival rate of 1800 pps (packets/second), a link capacity of 10 Mbps and a queue capacity of 100 packets. Compute the estimated values and the associated 90% confidence intervals of AM (in milliseconds) and PD, based on 100 runs of the simulator with $N = 10^4$. What do you conclude about the behaviour of the system concerning the queue capacity? The results should look like:

```
Exercise 9:
   AM = 3.5636 [3.4448 - 3.6825] msec
   PD = 0.0000 [0.0000 - 0.0000]
```

10. Consider the case of a packet arrival rate of 1800 pps and a link capacity of 10 Mbps. Compute the estimated values and the associated 90% confidence intervals of AM and PD based on 100 runs of the simulator with $N = 10^4$ considering:

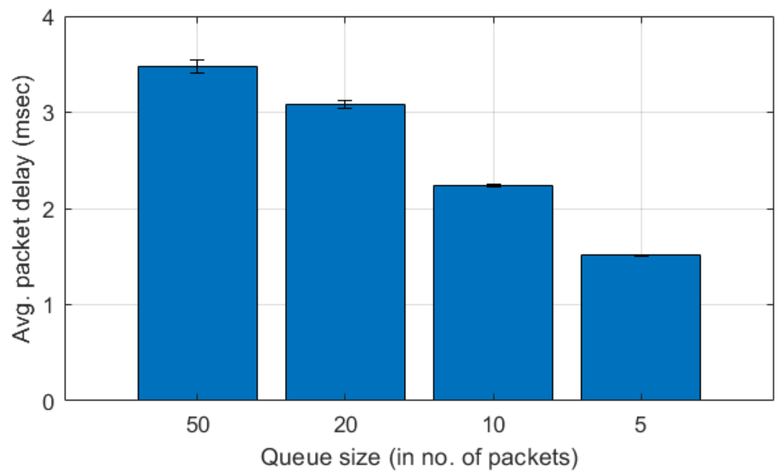
(a) a queue capacity of 50 packets,
(b) a queue capacity of 20 packets,
(c) a queue capacity of 10 packets,
(d) a queue capacity of 5 packets.

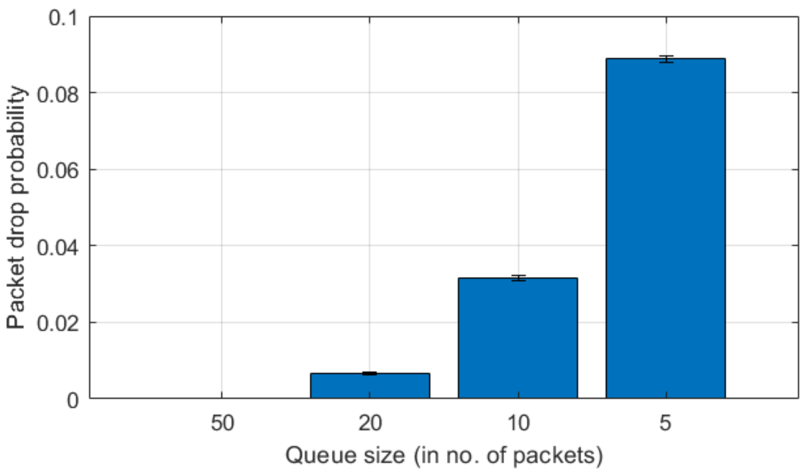
Footnote 1 https://www.mathworks.com/help/matlab/creating_plots/bar-chart-with-error-bars.html

Compare these results and draw conclusions. The results should look like:

```
Exercise 10:
   F= 50: AM = 3.4784 [3.4093 - 3.5475] msec
          PD = 0.0001 [0.0000 - 0.0001]
   F= 20: AM = 3.0843 [3.0451 - 3.1236] msec
          PD = 0.0067 [0.0063 - 0.0072]
   F= 10: AM = 2.2340 [2.2190 - 2.2489] msec
          PD = 0.0316 [0.0308 - 0.0324]
   F=  5: AM = 1.5132 [1.5085 - 1.5178] msec
          PD = 0.0888 [0.0880 - 0.0896]
```

11. Present the simulation results of **10.** in a bar chart with the confidence intervals in error bars on the same plot. Analyse the width of the obtained confidence intervals and take conclusions on the confidence of the obtained results. The visualization should look like:





12. For each case in **10.**, determine the theoretical values of the performance parameters AM and PD by modelling the data link connection as a $M/M/1/m$ queuing system. Compare these results with the simulation results (obtained in **10.**) and draw conclusions.